

TRANSACTIONS WITH REPLICATED DATA

1. Definition (Simple + Textbook)

- Transactions with replicated data means **a transaction that updates data which exists in multiple copies (replicas) and keeps all copies consistent.**
 - 👉 “Transaction = all steps done or none done”
 - 👉 “A group of steps that either fully happens or does not happen at all”
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👉 “Same data is stored in many places, so when we update, all copies must change correctly”

3. Basic Idea

- Data is stored in **multiple servers**
 - A transaction must update **all replicas**
 - All copies should show **same value**
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4. Example (VERY IMPORTANT)

👉 Bank account

- You withdraw money
- Balance must update in:
 - ATM
 - Mobile app
 - Bank server

✓ All must show same value

5. Key Points (WRITE IN EXAM)

- Data is replicated
- Transaction updates multiple copies
- Consistency must be maintained

- Requires coordination between nodes
 - Supports reliability
 - Avoids data mismatch
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6. Main Problems

- ✗ Some replicas updated, others not
 - ✗ Network delay causes inconsistency
 - ✗ Concurrent transactions conflict
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7. Requirements (ACID Properties)

- **Atomicity** → All replicas updated or none
 - **Consistency** → Same data everywhere
 - **Isolation** → Transactions don't interfere
 - **Durability** → Changes are permanent
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8. Methods (Important)

1. Primary-Based (Primary Copy)

- One server is main
 - All updates go through it
 - Then it updates others
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2. Active Replication

- All replicas execute same transaction

✓ Strong consistency

3. Quorum-Based

- Uses voting system

👉 Condition:
✓ $\text{Read} + \text{Write} > \text{Total replicas}$

9. Working (Steps)

1. Transaction starts
 2. Request sent to replicas
 3. Updates applied
 4. All replicas synchronized
 5. Commit done
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10. Diagram

Client



Transaction Manager



Replica1 Replica2 Replica3

11. Advantages

- High availability
 - Fault tolerance
 - Reliable system
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12. Disadvantages

- Complex
- High communication cost
- Synchronization overhead

5. ACID PROPERTIES

A – Atomicity

👉 “All or nothing”

- Either complete or cancel

C – Consistency

👉 “Correct data always”

- No invalid data

I – Isolation

👉 “No interference”

- Transactions don't affect each other

D – Durability

👉 “Permanent result”

- Data saved even after failure

6. Simple Diagram

Start → Operation1 → Operation2 → Commit ✓
OR
Abort ✗

7. Types of Results

- **Commit** → Success
- **Abort** → Failure